



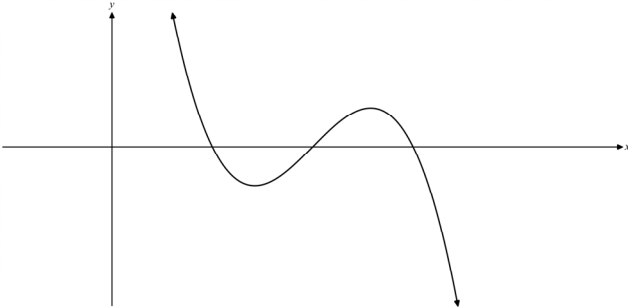
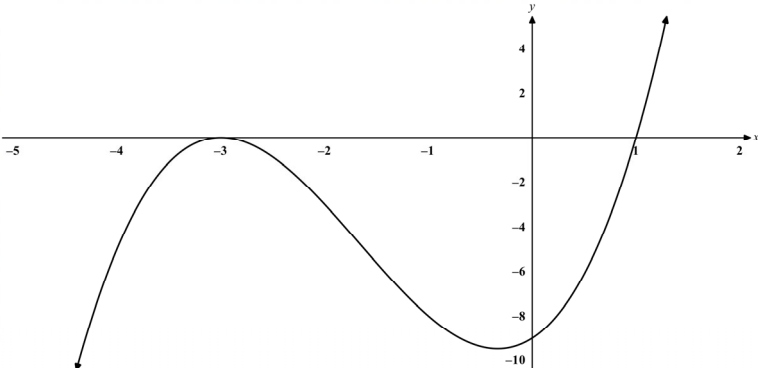
Year 11 Mathematics Extension Task 2

Thursday 1st August 2019

Time Allowed: 70 minutes plus 5 minutes reading time

Instructions:

- Write all your answers on the appropriate page in the answer booklet. Extra paper is available
- Show all necessary working.
- Marks may be deducted for careless or badly arranged work.
- Use black or blue pen to write your solutions. Black pen is preferred.

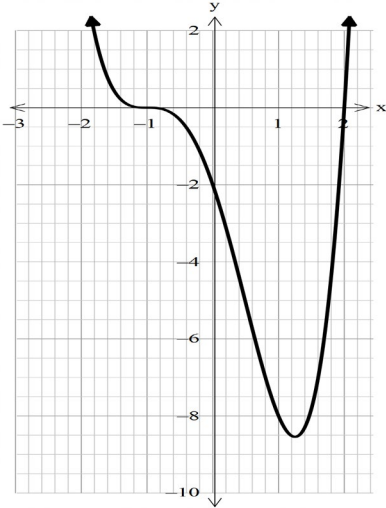
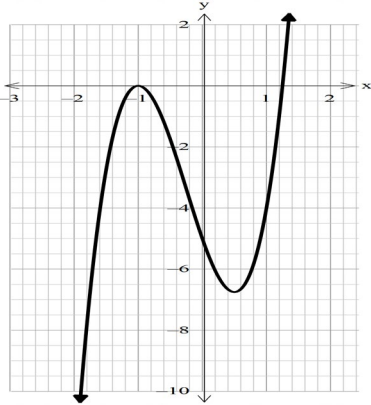
1.	Define the following terms that are used when describing polynomials; - coefficients - degree - leading term - monic polynomial - roots	1 1 1 1 1
2.	Consider the graph shown below  Give at least THREE reasons why the equation of the graph could NOT be $y = x^3 + 7x^2 - 11x$	3
3.	The two polynomials $P(x) = 2x^3 + 7x^2 - 3x + 12$ and $Q(x) = x^3 + 2x^2 - 6x + 21$ are solved simultaneously and the graph of $R(x) = P(x) - Q(x)$ is shown below  List at least THREE facts that this graph tells us about the points of intersection of $y = P(x)$ and $y = Q(x)$	3
4.	$P(x) = 5x^3 - 7x^2 + 8x - 3$ and $A(x) = x + 1$ - Using the division algorithm, divide $P(x)$ by $A(x)$ and write the result in the form $P(x) = A(x)Q(x) + R(x)$ - Use the remainder theorem to confirm your answer for $R(x)$	3 1

5.	(i) Use the factor theorem to find one factor of $P(x) = 6x^3 - 23x^2 + 11x + 12$	1
	(ii) Hence factorise $P(x)$ completely	2
6.	(i) On the axes provided in the answer booklet, sketch a monic polynomial that has a multiple root and a single root, neither of which is equal to zero, clearly labelling all intercepts.	3
	(ii) On the axes provided in the answer booklet, sketch the gradient function of your polynomial	3
	(iii) Give a possible equation for your polynomial function	1
	(iv) Using calculus, find the equation of your gradient function.	2
7.	$P(x)$ is a polynomial of degree 3. When $P(x)$ is divided by $(x + 3)$, the remainder is 2. When $P(x)$ is divided by $(x - 2)$, the remainder is -8. Find the remainder when $P(x)$ is divided by $x^2 + x - 6$	3
8.	α, β and γ are the roots of $x^3 + 2x^2 - 3x + 4 = 0$, find;	
	(i) $\alpha + \beta + \gamma$	1
	(ii) $\alpha\beta + \alpha\gamma + \beta\gamma$	1
	(iii) $\alpha\beta\gamma$	1
	(iv) $\alpha^2 + \beta^2 + \gamma^2$	2
	(v) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$	2
	(vi) $\alpha^3 + \beta^3 + \gamma^3$	2
	(vii) $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$	2
	(viii) a polynomial equation with roots α^2, β^2 and γ^2	2
9.	The cubic equation $x^3 - 15x + 4 = 0$ has three roots, two of which are reciprocals of each other.	
	(i) Find the third root	1
	(ii) Verify that the other two roots are reciprocals of each other	3
10.	If the equation $x^3 - 3kx + l = 0$ has a double root, prove that $l^2 = 4k^3$	3
11.	It is known that a polynomial has the following properties; • odd function • monic • coefficients are rational • degree is 5 • $P(1 + \sqrt{2}) = 0$ Find the polynomial that satisfies all of these properties. Give your answer in factored form, where no factor has a degree greater than 2.	3
END OF EXAM		

BAULKHAM HILLS HIGH SCHOOL

YEAR 11 MATHEMATICS EXTENSION ASSESSMENT TASK 2 2019 SOLUTIONS

Solution	Marks	Comments
1 (i) the coefficients of a polynomial are the constant multipliers of the algebraic terms	1	1 mark • Correct definition
1 (ii) the degree of a polynomial is the highest index	1	1 mark • Correct definition
1 (iii) the leading term of a polynomial is the algebraic term with the highest index	1	1 mark • Correct definition
1 (iv) a monic polynomial is a polynomial whose leading coefficient is equal to one	1	1 mark • Correct definition
1 (v) the roots are the solutions to the polynomial equation	1	1 mark • Correct definition
2. Some reasons would be; - The three roots are all positive, thus $\sum \alpha > 0$, this is not possible as the given polynomial has $\sum \alpha < 0$ - y intercept of the given polynomial is 0, yet the graph does not go through the origin - the leading coefficient of the given polynomial is positive, however the graph behaves like a polynomial whose leading coefficient is negative	3	1 mark for each correct reason - Reasons need to be unique, i.e. not the same reason expressed in a different way - Maximum of 3 marks
3. Some facts would be; - There are only two points of intersection - The graphs cut each other at $x = 1$ - The curves are tangential i.e. touch other, at $x = -3$	3	1 mark for each correct fact - Facts need to be unique, i.e. not the same fact expressed in a different way - Maximum of 3 marks
4 (i) $ \begin{array}{r} 5x^2 - 12x + 20 \\ x + 1 \overline{) 5x^3 - 7x^2 + 8x - 3} \\ \underline{5x^3 + 5x^2} \\ -12x^2 + 8x - 3 \\ \underline{-12x^2 - 12x} \\ 20x - 3 \\ \underline{20x + 20} \\ -23 \end{array} $ $\therefore 5x^3 - 7x^2 + 8x - 3 = (x + 1)(5x^2 - 12x + 20) - 23$	3	3 marks - Correct solution 2 marks - Completes the polynomial division 1 mark - Divides at least one term of $P(x)$ by $A(x)$
4 (ii) $R(x) = P(-1)$ $ \begin{aligned} &= 5(-1)^3 - 7(-1)^2 + 8(-1) - 3 \\ &= -5 - 7 - 8 - 3 \\ &= -23 \end{aligned} $	1	1 mark Correct solution, that includes the substitution step
5 (i) $P(3) = 6(3)^3 - 23(3)^2 + 11(3) + 12$ $ \begin{aligned} &= -162 - 207 + 33 + 12 \\ &= 0 \\ &\therefore (x - 3) \text{ is a factor} \end{aligned} $	1	1 mark - Substitutes a possible zero into the polynomial and obtains zero - Other possibilities are $P\left(-\frac{1}{2}\right)$ or $P\left(\frac{4}{3}\right)$
5 (ii) $P(x) = 6x^3 - 23x^2 + 11x + 12$ $ \begin{aligned} &= (x - 3)(6x^2 - 5x - 4) \\ &= (x - 3)(2x + 1)(3x - 4) \end{aligned} $	2	2 marks - Factorises completely 1 mark - Finds the other quadratic factor - Other possibilities are $3x^2 - 13x + 12$ or $2x^2 - 5x - 3$

Solution	Marks	Comments
6 (i) 	3	1 mark for each of these features • A curve with a labelled single root, not at $x = 0$ • A curve with a labelled multiple root, not at $x = 0$ • Correctly labels their y-intercept
6 (ii) 	3	1 mark for each of these features • Multiple root is reduced by an order of one • x-intercepts of gradient function match the stationary points of the polynomial function • Gradient function is positive/negative when the polynomial function is increasing/decreasing
6 (iii) $y = (x + 1)^3(x - 2)$	1	1 mark • An equation that correctly matches their sketch
6 (iv) $y = (x + 1)^3(x - 2)$ $\frac{dy}{dx} = (x + 1)^3(1) + (x - 2)[3(x + 1)^2(1)]$ $= (x + 1)^2(x + 1 + 3x - 6)$ $= (x + 1)^2(4x - 5)$	2	2 marks • Gradient function matches the graph drawn 1 mark • Finds the gradient function
7. As the divisor is a quadratic, then the remainder is a linear function or a constant, i.e. $R(x) = ax + b$ $\therefore P(x) = (x^2 + x - 6)Q(x) + ax + b$ $= (x + 3)(x - 2)Q(x) + ax + b$ $\begin{array}{rcl} P(-3) = 2 & & P(2) = -8 \\ -3a + b = 2 & & 2a + b = -8 \\ -3a + b = 2 & & \\ 2a + b = -8 & & \\ -5a & = & 10 \\ a = -2 & \therefore & b = -4 \end{array}$ thus the remainder is $-2x - 4$	3	3 marks • Correct solution 2 marks • Uses $P(2)$ and $P(-3)$ to find two simultaneous equations 1 mark • Establishing the remainder could be a linear function
8 (i) $\alpha + \beta + \gamma = -2$	1	1 mark • Correct answer
8 (ii) $\alpha\beta + \alpha\gamma + \beta\gamma = -3$	1	1 mark • Correct answer
8 (iii) $\alpha\beta\gamma = -4$	1	1 mark • Correct answer

Solution	Marks	Comments
8 (iv) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $= (-2)^2 - 2(-3)$ $= 4 + 6$ $= 10$	2	2 marks • Correct solution 1 mark • Rearranges the perfect square to obtain a formula for $\alpha^2 + \beta^2 + \gamma^2$
8 (v) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$ $= \frac{-3}{-4}$ $= \frac{3}{4}$	2	2 marks • Correct solution 1 mark • Forms a single fraction that previous answers can be substituted into
8 (vi) If α , β and γ are the roots of $x^3 + 2x^2 - 3x + 4 = 0$, then; $\alpha^3 + 2\alpha^2 - 3\alpha + 4 = 0$ $\beta^3 + 2\beta^2 - 3\beta + 4 = 0$ $\gamma^3 + 2\gamma^2 - 3\gamma + 4 = 0$ $\alpha^3 + \beta^3 + \gamma^3 + 2(\alpha^2 + \beta^2 + \gamma^2) - 3(\alpha + \beta + \gamma) + 12 = 0$ $\alpha^3 + \beta^3 + \gamma^3 = -2(10) + 3(-2) - 12$ $= -38$	2	2 marks • Correct solution 1 mark • Substitutes α , β and γ into the polynomial, or equivalent • rearranges the perfect cube to obtain a formula for $\alpha^3 + \beta^3 + \gamma^3$
8 (vii) $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$ $= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2)$ $= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$ $= (-3)^2 - 2(-4)(-2)$ $= 9 - 16$ $= -7$	2	2 marks • Correct solution 1 mark • Uses the $\Sigma\alpha^2$ formula to create an expression to substitute known answers into.
8 (viii) $\alpha^2 + \beta^2 + \gamma^2 = 10$ $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = -7$ $\alpha^2\beta^2\gamma^2 = (-4)^2 = 16$ $\therefore \text{equation is } x^3 - 10x^2 - 7x - 16 = 0$	2	2 marks • Correct solution 1 mark • evaluates $\alpha^2\beta^2\gamma^2$ • uses $\Sigma\alpha^2$ and $\Sigma\alpha^2\beta^2$ to create a polynomial
9 (i) $x^3 - 15x + 4 = 0$ Roots are $\alpha, \frac{1}{\alpha}$ and β Using product of roots $(\alpha)\left(\frac{1}{\alpha}\right)(\beta) = -4$ $\beta = -4$	1	1 mark • Correct answer

Solution	Marks	Comments
9 (ii) using sum of roots $\alpha + \frac{1}{\alpha} + \beta = 0$ $\alpha + \frac{1}{\alpha} - 4 = 0$ $\alpha^2 - 4\alpha + 1 = 0$ $\alpha = \frac{4 \pm \sqrt{16 - 4}}{2}$ $= \frac{4 \pm \sqrt{12}}{2}$ $= 2 \pm \sqrt{3}$ Product of the other two roots $= (2 + \sqrt{3})(2 - \sqrt{3})$ $= 4 - 3$ $= 1$ Since the product of the two roots is one, they must be reciprocals of each other.	3	3 marks • Correct solution 2 marks • Finds the other two roots 1 mark • Finds the other quadratic factor
10. $P(x) = x^3 - 3kx + l$ $P'(x) = 0$ $P'(x) = 3x^2 - 3k = 0$ $x^2 = k$ $x = \pm\sqrt{k}$ $P(x) = 0$ $(\pm\sqrt{k})^3 - 3k(\pm\sqrt{k}) + l = 0$ $\pm k\sqrt{k} \mp 3k\sqrt{k} + l = 0$ $l = \pm 2k\sqrt{k}$ $l^2 = 4k^2 \times k$ $l^2 = 4k^3$	3	3 marks • Correct solution 2 marks • Finds the possible double roots 1 mark • Uses the fact that $P'(x) = 0$
11. All odd functions have rotational symmetry about the origin, thus $f(-x) = -f(x)$ $P(0) = -P(0)$ $2 \times P(0) = 0$ $P(0) = 0 \Rightarrow x \text{ is a factor}$ As all of the coefficients of the polynomial are real; If $-1 - \sqrt{2}$ is a root, its conjugate must also be a root By rotational symmetry $1 + \sqrt{2}$ is also a root, and its conjugate must also be a root. The five roots are $0, 1 \pm \sqrt{2}, -1 \pm \sqrt{2}$ $\alpha = 1 + \sqrt{2} \text{ and } \beta = 1 - \sqrt{2} \quad \gamma = -1 + \sqrt{2} \text{ and } \delta = -1 - \sqrt{2}$ $\alpha + \beta = 2 \quad \gamma + \delta = -2$ $\alpha\beta = -1 \quad \gamma\delta = -1$ $\therefore x^2 - 2x - 1$ is a factor $\therefore x^2 + 2x - 3$ is a factor Thus $P(x) = x(x^2 - 2x - 1)(x^2 + 2x - 1)$	3	3 marks • Correct solution 2 marks • Finds a quadratic factor • Establishes two of the points below 1 mark • Establishes that x is a factor • Establishes $1 - \sqrt{2}$ i.e. conjugate surd is a root • Establishes $\frac{1}{1 + \sqrt{2}}$ i.e. reciprocal is a root • Uses the property of odd functions to establish $P(-1 - \sqrt{2}) = 0$ i.e. $1 + \sqrt{2}$ is a root